



COMPUTER SECURITY

2025

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(Solved)



SENIOR

25

BY AB & AE & AM & MR

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Choose the correct answer:

1.	The need for <u>keeping information secret</u> .				
	a) Confidentiality	b) Integrity	c) Availability	d) None	
2.	Trustworthiness of data and resources.				
	a) Confidentiality	b) Integrity	c) Availability	d) None	
3.	The ability to use the information or resource				
	a) Confidentiality	b) Integrity	c) Availability	d) None	
4.	Timely response is an expectation of _____ security goal				
	a) Integrity	b) Availability	c) Confidentiality	d) All are true	
5.	_____ means that some unauthorized party (person, program) <u>has gained access to data</u>				
	a) Interception	b) Modification	c) Interruption	d) Fabrication	
6.	_____ an asset of the system <i>becomes lost, unavailable or unusable</i>				
	a) Interruption	b) Interception	c) Modification	d) Fabrication	
7.	_____ means that some unauthorized party (person, program) has <u><i>gained access to an asset</i></u>				
	a) Interruption	b) Interception	c) Modification	d) Fabrication	
8.	_____ an additional computation, or modifying data being transmitted by the network.				
	a) Interruption	b) Interception	c) Modification	d) Fabrication	
9.	_____ counterfeit objects on a computing system. (adding records to an existing data base or insertion of spurious transactions to a network communication system.				
	a) Interruption	b) Interception	c) Modification	d) Fabrication	
10.	Cryptography is used as a defense technique against the following attacks except _____				
	a) Modification	b) Interception	c) Interruption	d) Fabrication	
11.	Erasure of program or data or malfunction of an OS file manager is considered _____				
	a) Modification	b) Interception	c) Interruption	d) Fabrication	
12.	Flooding is an attack of _____				
	a) Integrity	b) Availability	c) Confidentiality	d) All are true	
13.	Spoofing is an attack of _____				
	a) Integrity	b) Availability	c) Confidentiality	d) All are true	
14.	_____ is when <u>one host pretends to be another</u>				
	a) Eavesdrop	b) Spoofing	c) Wiretap	d) Session Hijacking	
15.	A sender <u>should not be able falsely to deny later that he/she sent a message</u>				
	a) Integrity	b) Confidentiality	c) Privacy	d) Non-repudiation	
16.	Program controls include:				
	a) Peer review	b) Proof of program correctness	c) Independent testing	d) All	



17.	Legal and ethical controls are classified as:							
	a)	Hardware controls	b)	Software controls	c)	Policy controls	d)	Physical controls
18.	Cyber security rules for daily practicing are classified as:							
	a)	Hardware controls	b)	Software controls	c)	Administration controls	d)	Physical controls
19.	Eavesdropping and impersonate are attacks of _____ Respectively							
	a)	Availability , Confidentiality	b)	Availability , Integrity	c)	Confidentiality , Integrity	d)	Integrity , Confidentiality
20.	Choose the correct sentence?							
	a)	Erasure of a program is considered an interruption threat						
	b)	TOCTOU is considered an attack of availability						
	c)	Flooding are attacks of confidentiality						
	d)	Error is a mistake in interpreting a syntax error in a piece of code						
21.	_____ Consuming processor capacity, system memory, disk space							
	a)	Bacteria	b)	Resident	c)	Virus	d)	Worm
22.	_____ Attaches itself to program and propagates copies of itself to other programs							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
23.	_____ Contains unexpected, additional functionality							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
24.	_____ Triggers action when condition occurs							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
25.	_____ Triggers action when specified time occurs							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
26.	_____ Allows unauthorized access to functionality							
	a)	Virus	b)	Trapdoor	c)	Worm	d)	Rabbit
27.	_____ Propagates copies of itself through a network							
	a)	Virus	b)	Trapdoor	c)	Worm	d)	Rabbit
28.	_____ Replicates itself without limit to exhaust resources							
	a)	Virus	b)	Trapdoor	c)	Worm	d)	Rabbit
29.	_____ is a class of malicious code that "detonates" or goes off when a specified condition occurs							
	a)	Virus	b)	Trojan horse	c)	Logic bomb	d)	Time bomb
30.	The _____ is a code embedded in program that is set to explode when certain condition is/are met							
	a)	Trojan house	b)	Trapdoor	c)	Worm	d)	Logical bomb
31.	_____ replicates a copy of itself to other systems to execute a malicious code							
	a)	Virus	b)	Worm	c)	Trojan horse	d)	Trapdoor



32.	_____ is a program that spreads copies of itself through a network				
	a) Virus	b) Trapdoor	c) Worm	d) Rabbit	
33.	_____ is a feature in a program by which someone can access the program <u>other than by the obvious, direct call</u> , perhaps with special privileges				
	a) Virus	b) Trapdoor	c) Worm	d) Rabbit	
34.	_____ virus runs when its attached program executes and terminates when its attached program ends				
	a) Transient	b) Resident	c) Virus	d) Worm	
35.	_____ virus locates itself in memory so that it can <u>remain active</u> even after its attached program ends				
	a) Transient	b) Resident	c) Virus	d) Worm	
36.	A type of virus that executes only once , spreading its infection and causing its effect in that one execution .				
	a) Memory resident virus	b) Boot sector virus	c) One time execution	d) Appended virus	
37.	A kind of malicious software runs when a computer is started , it changes control on the OS instructions				
	a) Memory resident virus	b) Boot sector virus	c) One time execution	d) Appended virus	
38.	A type of virus which is active as long as the computer running				
	a) Memory resident virus	b) Brain virus	c) Polymorphic virus	d) Logic bomb	
39.	Virus is active as long as the computer running				
	a) Memory resident	b) Brain virus	c) Polymorphic virus	d) Trojan horse	
40.	Viruses resident in application programs and libraries are categorized as:				
	a) One time execution virus	b) Boot sector virus	c) Other homes of virus	d) Memory resident	
41.	A program virus attaches itself to a program ; then, whenever the program is run, the virus is activated. This kind of attachment is usually known as _____				
	a) integrated virus	b) Surrounded virus	c) Appended virus	d) replacement virus	
42.	controls against program threats include the following actions <u>except</u> _____				
	a) peer review process	b) Independent testing	c) Proof of correctness	d) Decapsulation of different process	
43.	Each virus has a signature that could be used to detect its existence, such as:				
	a) Its location	b) its execution performance	c) its replication technique	d) All	
44.	What is a weakness of Caesar cipher?				
	a) Vulnerable to brute-force attacks	b) Vulnerable to frequency analysis	c) limited key space	d) all	



45.	What is true about Steganography?							
	a)	it is used to encrypt images and send it in unreadable forms						
	b)	The encrypted file has smaller size compared with the original one						
	c)	it can be broken by brute-force attack						
	d)	it uses images to hide important data in LSB of each image's pixel						
46.	Steganography uses _____ in graphic image to hide important data							
	a)	LSB	b)	HSB	c)	MSB	d)	NSB
47.	_____ is a mistake in <i>interpreting a requirement</i> or a syntax error in a piece of code							
	a)	Error	b)	Flaws	c)	Bugs	d)	Fault
48.	_____ is a human mistake in performing some software activity that <u>may lead to a fault</u> in computer program							
	a)	Error	b)	Flaws	c)	Bugs	d)	Fault
49.	Validation error , domain error , and boundary condition violation are types of _____							
	a)	Error	b)	Flaws	c)	Bugs	d)	Fault
50.	A _____ <u>may cause a failure</u> (which is a departure from the system's required behavior).							
	a)	Bug	b)	Error	c)	fault	d)	failure
51.	A _____ is an inside view of the system , <u>seen by the developers</u> , whereas a _____ is an outside view <u>seen by the user</u>							
	a)	Failure, fault	b)	Fault, Failure	c)	Error, Bug	d)	Fault, Flaws
52.	What is true about program <u>flaws</u> ?							
	a)	It could be validation error or domain error						
	b)	It could lead to failure						
	c)	It could be malicious or non malicious						
	d)	All are true						
53.	Consist of replacing the valid source and/or destination IP address with false ones							
	a)	Smurf attack	b)	Spoofing attack	c)	DoS	d)	Sniffing attack
54.	For exchanging symmetric key among 10 user , we need _____ exchange operation							
	$n = 10, \frac{n \times n - 1}{2} = \frac{10 \times 9}{2} = 45$							
	a)	45	b)	100	c)	90	d)	1000
55.	Authentication based on <u>Something the individual has</u> as:							
	a)	ID	b)	Password	c)	ATM card	d)	PIN
56.	Authentication based on <u>Something the individual knows</u> as:							
	a)	PIN	b)	Password	c)	ATM card	d)	A & B
57.	Authentication based on <u>Something the individual is</u> as:							
	a)	Voice patterns	b)	Handwriting	c)	Fingerprint	d)	All



58.	Password and PIN are examples for user authentication of something _____			
	a) The user has	b) The user is	c) The user knows	d) The user
59.	What is the Row Transposition Cipher?			
	a) A substitution cipher that rearranges the order of letters in a message			
	b) A cipher that replaces each letter in a message with a numerical value			
	c) A cipher that applies a mathematical formula to each character in a message			
	d) A cipher that rearranges the order of words in a message			
60.	Which of the following is a common technique used with transposition cipher?			
	a) Rail fence	b) Caesar	c) Vagenre	d) Playfair
61.	The mechanisms used to confirm a user's identity is known as:			
	a) Identification	b) Authentication	c) Authorization	d) Access limitation
62.	Each virus has a signature that could be used to detect its existence, such as:			
	a) Location	b) Execution performance	c) Spreading	d) All are true
63.	In _____, Users interact with directory to obtain any desired public key securely			
	a) Certificate authority	b) Public Key authority	c) Public announcement	d) Public Discovery
64.	_____ allow key exchange without real-time access to public-key authority			
	a) Certificate authority	b) Public key authority	c) Public announcement	d) Public directory
65.	Example for classical substitution encryption			
	a) Playfair	b) DES	c) 3DES	d) AES
66.	Triple-DES with two keys is defined by $C = EK_3(DK_2(EK_1(P)))$, if K_1 equal K_3 , the triple-DES can be considered هنا هو بيقولك ان k_1 يساوي k_3 , وأول خطوه حصلت انه شفر ال P ب k_1 وبعدين ثاني خطوة فك تشفيرها ب k_2 بس ده مش هيرجع لك P الاصلية لان k_1 مش هي k_2 فدي هتعتبر خطوة تشفير تانية، وبعدين قام مشفر ب k_3 فكذا حصل تشفير ثلاث مرات يبقى ده triple DES			
	a) Double DES	b) Triple-DES	c) Single-DES	d) Useless encryption
67.	Triple-DES with two keys is defined by $C = Ek_1(Dk_2(Ek_1(P)))$, if K_1 equals K_2 , the triple-DES will be هنا هو بيقولك ان k_1 يساوي k_2 , وأول خطوه حصلت انه شفر ال P ب k_1 وبعدين ثاني خطوة فك تشفيرها ب k_2 فهيرجع ال P الأصلية عادي يعني اول خطوه اتلغت بتاني خطوة، وبعدين رجع شفر ب k_1 فيبقى كأنه عمل تشفير مره واحده بس فهتبقى single DES			
	a) Double DES	b) Triple-DES	c) Single-DES	d) Useless encryption
68.	Which of the following is a good description for Triple-DES?			
	a) $C = Ek_1(Dk_2(Ek_1(P)))$	b) $C = Ek_2(Dk_1(Ek_1(P)))$	c) $C = Ek_3(Ek_2(Ek_1(P)))$	d) $C = Dk_3(Dk_2(Dk_1(P)))$
69.	DES encrypt data in blocks of size _____, using a key of length _____			
	a) 64-bit , 32-bit	b) 32-bit, 32-bit	c) 32-bit , 56-bit	d) 64-bit , 56-bit
70.	What is the number of rounds in single DES?			
	a) 64	b) 56	c) 48	d) 16



71.	What is the block size of <u>triple DES</u> ?			
	a) 64	b) 128	c) 256	d) 512
72.	_____ is an attempt to discover PW by attempting to use every possible PW from a predefined list.			
	a) Brute-force attack	b) Dictionary attack	c) Sniffing attack	d) Cracking
73.	_____ provides a centralized authentication server to authenticate users to servers and servers to users .			
	a) Trusted Third Party (TTP)	b) Trusted Peer Party (TPP)	c) Trusted Party Model (TPM)	d) None
74.	_____ is a protocol that <u>produces the same effect as a real signature</u>			
	a) Digital signatures	b) Symmetric Key	c) Asymmetric Key	d) None
75.	Digital signatures provide the ability to:			
	a) verify author, date & time of signature	b) be verified by third parties to resolve disputes	c) authenticate message contents	d) All are true
76.	_____ refers to a key storage facility which provides secure storage of keys for future use, e.g.			
	a) Storage of Keying Material	b) Key Archival	c) Key Replacement	d) Key Recovery
77.	_____ refers to procedures by which keys for notarization or nonrepudiation services can be securely archived .			
	a) Storage of Keying Material	b) Key Archival	c) Key Replacement	d) Key Recovery
78.	_____ enables parties to update securely their keying material.			
	a) Storage of Keying Material	b) Key Archival	c) Key Replacement	d) Key Recovery
79.	_____ refers to cryptographic keys which may become lost due to human error, software bugs, or hardware malfunction			
	a) Storage of Keying Material	b) Key Archival	c) Key Replacement	d) Key Recovery
80.	_____ refers to procedures by which parties are assured of the secure destruction of keys that are no longer needed			
	a) Storage of Keying Material	b) Key Archival	c) Key Replacement	d) Key Deletion
81.	_____ refers to the procedure by which keys or pairs of keys of good cryptographic quality are securely and unpredictably generated			
	a) Storage of Keying Material	b) Key Archival	c) Key Replacement	d) Key Generation
82.	_____ provide some link between an entity and its uniquely defined keys			
	a) Entity Registration	b) Key Replacement	c) Entity Authentication	d) Entity Replacement
83.	_____ is to corroborate that an entity is the one claimed			
	a) Key Replacement	c) Entity Authentication	d) Entity Replacement	d) Entity Registration



84.	_____ is the registration of messages to attest at a later stage its content origin .			
	a) Notarization	b) Proof	c) Nonrepudiation of Reception	d) none
85.	_____ secret keys and possibly other data are to be kept confidential while transmitted or stored			
	a) Data Confidentiality	b) Modification Detection	c) Replay Detection	d) Timeliness
86.	_____ is to counter the active threat of unauthorized modification of data items			
	a) Data Confidentiality	b) Modification Detection	c) Replay Detection	d) Timeliness
87.	_____ is to counter unauthorized duplication of data items.			
	a) Data Confidentiality	b) Modification Detection	c) Replay Detection	d) Timeliness
88.	_____ is required that the response to a challenge message is prompt and does not allow for playback of some authentic response message by an impersonator.			
	a) Data Confidentiality	b) Modification Detection	c) Replay Detection	d) Timeliness
89.	_____ It involves storing, safeguarding, and activating keys			
	a) key management	b) key distribution	c) Key Replacement	d) Key Recovery
90.	_____ refers to the procedures by which keys are securely provided to parties legitimately asking for them .			
	a) key management	b) key distribution	c) Key Replacement	d) Key Recovery
91.	The _____ private key is used to decrypt cipher text in asymmetric cryptosystem			
	a) Sender	b) Receiver	c) Both A & B	d) None
92.	The _____ key is used to encrypt plain text in asymmetric cryptosystem			
	a) public	b) private	c) Both A & B	d) None
93.	_____ is the science of code breaking in a cryptographic schema			
	a) Cryptography	b) Cryptanalysis	c) Cryptology	d) None
94.	_____ is the study of encryption principles and methods.			
	a) Cryptography	b) Cryptanalysis	c) Cryptology	d) None
95.	_____ is the field of both cryptography and cryptanalysis.			
	a) Cryptography	b) Cryptanalysis	c) Cryptology	d) None
96.	In cryptography system, _____ is the original message before being transmitted			
	a) Cipher text	b) encrypted	c) Plain text	d) None
97.	In Caesar cipher, shifting a letter by +3 positions means that			
	a) A becomes D, B becomes E, C becomes F, and so on			
	b) A becomes X, B becomes Y, C becomes Z, and so on			
	c) A becomes C, B becomes F, C becomes I, and so on			
	d) A becomes B, B becomes C, C becomes D, and so on			



True/False:

1.	<u>Spoofing</u> attack consist of <u>replacing the valid</u> source and/or destination IP address <u>with false one</u>	T
2.	<u>Bug</u> is a <u>mistake in interpreting a requirement</u> or <u>syntax error in a piece of code</u>	T
3.	For exchanging symmetric key among 10 users, we need 100 exchange operation (45)	F
4.	Erasure of program or data or malfunction of an OS file manager is considered fabrication (interruption)	F
5.	<u>Validation error</u> , <u>domain error</u> , and <u>boundary condition violation</u> are types of <u>flaws</u>	T
6.	Erasure of program or data or malfunction of an OS file manager is considered Modification (interruption)	F
7.	<u>Timely response</u> is an expectation of <u>availability</u> security goal.	T
8.	<u>Verification of the user's identity</u> is known as authorization process (authentication)	F
9.	Digital signatures provide the ability to verify both owner and content of message	T
10.	The plain text must be broken into a block size of 128bit as an input of Triple DES (64-bit)	F
11.	<u>Polymorphic virus</u> is a form of virus explicitly designed to <u>hide itself from detection</u>	T
12.	A <u>boot sector</u> virus changes control on the OS instructions, when a computer <u>is started</u>	T
13.	Playfair cipher rearranges the order of words in a message (letters)	F
14.	Shift cipher applies a <u>mathematical formula</u> to each letter in a message that replaces each letter with a numerical value. (Vingere)	F
15.	<u>Flooding</u> is an attack of <u>availability</u>	T
16.	Authorization is mechanism used to confirm a user's identity (authentication)	F
17.	Fingerprint is used to verify the user as an example for something <u>user has</u> (ATM card)	F
18.	Trapdoor is a malicious software runs when a computer is started, it Changes control on the OS instructions (boot sector virus)	F
19.	Program controls must include <u>peer review steps</u>	T
20.	In context of Program controls, <u>proof of program correctness</u> is done based on <u>check list</u>	T
21.	<u>Code flaws</u> include validation error or domain error	T

